CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 4: Project Planning Phase

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 4: Project Planning Phase**
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Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Work Breakdown Structure (WBS) & Kanban Task Mapping

The project execution strictly follows the **8 Epics and 20 Tasks** defined in the **SmartBridge MySkillWallet** capstone curriculum. All 20 cards have been mapped and completed:

Epic	Epic Title	Scope	Tasks Executed	Status
Epic 1	**Problem Definition & Data Collection**	4 Tasks	1.1 Business Problem, 1.2 Requirements, 1.3 Data Collection, 1.4 Dataset Structure	**100% Completed**
Epic 2	**Data Architecture Setup**	3 Tasks	2.1 Warehouse & DB, 2.2 Schemas & Stage, 2.3 CSV File Format	**100% Completed**
Epic 3	**Data Ingestion**	2 Tasks	3.1 Create RAW Tables, 3.2 Loading Tables & Validating	**100% Completed**
Epic 4	**Data Preparation & Modelling**	2 Tasks	4.1 Staging Cleansing, 4.2 Curated Star Schema & Views	**100% Completed**
Epic 5	**Business Analytics**	2 Tasks	5.1 Designing KPIs, 5.2 Designing Visuals & Queries	**100% Completed**
Epic 6	**Streamlit Application Development**	1 Task	6.1 Interactive Multi-Tab Dashboard Deployment	**100% Completed**
Epic 7	**Performance Testing**	4 Tasks	7.1 Data Volume, 7.2 Filter Efficiency, 7.3 New Fields, 7.4 Query History	**100% Completed**
Epic 8	**Demonstration & Documentation**	2 Tasks	8.1 Video Walkthrough (4m 37s), 8.2 Comprehensive Project Report	**100% Completed**

2. Project Sprint Schedule & Timeline

Sprint Window	Focus Area	Milestones Achieved
Sprint 1 (Days 1–3)	**Requirements & Architecture**	Conducted stakeholder interviews for Priya, Daniel, and Grace. Configured ORDERPULSE_WH and 3 Medallion schemas.
Sprint 2 (Days 4–6)	**Ingestion & Staging Pipeline**	Staged 47,113 raw rows via internal stages. Built SQL transformation pipeline casting types and deriving business flags.
Sprint 3 (Days 7–9)	**Curated Star Schema & Marts**	Constructed FACT_ORDERS, FACT_ORDER_ITEMS, and 5 Dimensions with MD5 surrogate keys. Implemented persona views.
Sprint 4 (Days 10–12)	**Streamlit Dashboard & UI**	Engineered 5-tab Streamlit application with interactive Plotly visualizations, KPI summary cards, and dynamic sidebar filters.
Sprint 5 (Days 13–14)	**Performance Testing & Audit**	Conducted partition pruning benchmarks (6.5x speedup), audited 24 engineered fields, recorded 1080p demo video, and submitted.

3. Risk Management & Mitigation Strategies

Identified Technical Risk	Impact	Proactive Mitigation Architecture
Snowflake Credit Exhaustion	High	Configured warehouse auto-suspend strictly to 60 seconds and auto-resume to TRUE; executed all heavy test cycles on X-Small warehouse.
Missing / Corrupted Source Data	Medium	Staged raw CSVs using ON_ERROR = 'CONTINUE'; validated exact row counts against source files post-load.
Dashboard Query Latency	Medium	Pre-computed RFM tiers and meal periods into Curated tables; used Result Cache and partition pruning for sub-second page refreshes.